

AP CALCULUS AB - UNIT 4

Guided Notes 4.6: L'Hopital's Rule



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we recognize the indeterminate quotient forms $0/0$ and ∞/∞ ?

- Recognize the indeterminate quotient forms $0/0$ and ∞/∞ .
- Verify the hypotheses of L'Hopital's Rule before differentiating numerator and denominator.
- Evaluate indeterminate limits, including repeated applications when justified.
- Rewrite selected products, differences, and powers into allowed quotient forms.

Key Knowledge, Vocabulary, and Notation

Allowed formsDirect substitution must give $0/0$ or ∞/∞ .**L'Hopital's Rule**If the hypotheses hold, $\lim \frac{f}{g} = \lim \frac{f'}{g'}$ when the derivative ratio limit exists or is infinite.**Not a quotient rule**

Differentiate numerator and denominator separately; do not differentiate the quotient by the Quotient Rule.

Repeated use

Recheck the form after each application before applying the rule again.

Other formsProducts $0 \cdot \infty$, differences $\infty - \infty$, and powers require algebraic or logarithmic transformation first.**Alternative methods**

Factoring, rationalizing, known limits, or direct substitution may be simpler and should be considered first.

Concept Development and Strategy

1. Always show the substitution that produces an allowed indeterminate form.
2. Differentiate the numerator and denominator as separate functions and then reevaluate the resulting limit.
3. A form such as $1/0$, $0/\infty$, or $\infty/0$ is not indeterminate and does not justify the rule.
4. For 1^∞ , 0^0 , or ∞^0 , take logarithms, evaluate the logarithmic product, and exponentiate at the end.
5. State when repeated application is justified rather than applying derivatives mechanically.

Worked Example: One application

Evaluate $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x}$.**Answer and Reasoning**The form is $0/0$. Differentiate top and bottom: $\lim 2e^{2x} = 2$.

Worked Example: Repeated application

Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$.

Answer and Reasoning

It is $0/0$. First application gives $\lim \sin x / (2x)$, still $0/0$. A second gives $\lim \cos x / 2 = 1/2$.

Worked Example: Infinity over infinity

Evaluate $\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}}$.

Answer and Reasoning

The form is ∞/∞ . The derivative ratio is $(1/x)/(1/(2\sqrt{x})) = 2/\sqrt{x} \rightarrow 0$.

Worked Example: Transform a product

Evaluate $\lim_{x \rightarrow 0^+} x \ln x$.

Answer and Reasoning

Rewrite as $\ln x / (1/x)$, an $-\infty/\infty$ form. The derivative ratio is $(1/x)/(-1/x^2) = -x \rightarrow 0$.

Guided Practice and Discussion

Directions

Show the setup, preserve exact values unless a decimal is requested, and include units and theorem conditions whenever the context requires them.

1. Identify the form of $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

2. Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$.

3. Evaluate $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$.

4. Evaluate $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$.

5. Evaluate $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$.

6. Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$.

7. Suppose $f(0) = g(0) = 0$, $g'(0) \neq 0$, and both are differentiable near 0. State a condition guaranteeing $\lim f/g = f'(0)/g'(0)$.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer, then identify the mathematical evidence supporting your choice.

1. L'Hopital's Rule may be applied directly when substitution gives

(A) $3/0$

(B) $0/5$

(C) $0/0$

(D) $\infty - \infty$

2. $\lim_{x \rightarrow 0} (e^{4x} - 1)/x$ equals

(A) 1

(B) 4

(C) e^4

(D) 0

3. $\lim_{x \rightarrow 0} \frac{\sin(2x) - 2x}{x^3}$ is

(A) $-4/3$

(B) $-2/3$

(C) 0

(D) $4/3$

Common Errors and AP Precision

- State the quantity, sign, input value, and units in every contextual interpretation.
- In related rates, differentiate before substituting values. In linearization, identify the base point. For L'Hopital's Rule, verify $0/0$ or ∞/∞ first.
- A complete AP response names the definition, sign change, theorem, or comparison that supports the conclusion.

Exit Ticket

Evaluate $\lim_{x \rightarrow 0^+} (\sin x)^x$.

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